
From Attack Success Rate to Reachability: Actionable Distributional Diagnostics for LLM Safety

Anonymous authors

Paper under double-blind review

Abstract

Jailbreak benchmarks report attack success rate (ASR): did a sampled response violate a safety policy? This outcome-level view is necessary but incomplete. Autoregressive language models define *distributions* over continuations; a greedy refusal can hide substantial probability mass on nearby harmful branches. We propose *bounded continuation analysis* (BCA), an interpretability method that exposes this local output structure by constructing an α - k bounded continuation tree, labeling leaves with a semantic harm judge, and computing finite-horizon reachability $\mathbb{P}[\mathbf{F} \text{ harm}]$. We show how this distributional reading becomes *actionable*: it triages models with matched ASR, ranks jailbreak templates before spending red-team queries, predicts empirical attack success under independent judges (Spearman $\rho \in [0.53, 0.73]$), and—when integrated into adaptive red-teaming—can expand jailbreak *coverage* via complementary search modes, though attacker-objective gains from BCA pruning are judge- and seed-conditional. Our framing aligns with the Actionable Interpretability agenda: micro-level structure in the generation distribution, translated into deployment diagnostics rather than post-hoc explanation alone.

1 Introduction

Large language model (LLM) safety is often measured as a binary event: given a harmful behavior and an adversarial prompt, does the model’s response count as a jailbreak? HarmBench (Mazeika et al., 2024) and JailbreakBench (Chao et al., 2024) standardize behaviors, threat models, and graders, making ASR comparisons more reproducible. Yet ASR is fundamentally a *sampled surface* statistic. Two models can refuse on the modal decode while differing sharply in how much of their conditional output mass routes toward compliance (Chouldechova et al., 2026).

This gap is interpretability with teeth. An autoregressive model induces a probability tree over token prefixes. *Where* probability mass sits—on refusal markers, hedging, or direct compliance—is a micro-level property of the generation mechanism. If we can estimate that structure, safety teams can act on it: prioritize red-team budgets, compare models before deployment, and generate diverse adversarial training data.

We operationalize this via **bounded continuation analysis** (BCA). Given prompt x , target model M , horizon L , and bounding parameters (α, k) , BCA expands the likely branches of M ’s next-token distribution, labels terminal continuations with a semantic harm judge J , and computes

$$H_{\alpha,k,L}(x; M, J) = \mathbb{P}_{\tilde{P}}[\mathbf{F} \text{ harm}],$$

the probability of eventually reaching a judge-labeled harmful leaf in the bounded process $\tilde{P}$. When the tree is small, we compute this exactly by sparse backward induction on the induced finite DTMC (Gross et al., 2025; Baier & Katoen, 2008); otherwise we estimate it by statistical model checking (SMC) with confidence intervals (Younes & Simmons, 2002; Hoeffding, 1963).

We package bounded reachability into **generative-attack susceptibility** (GAS), a query-budgeted family that nests ASR, BCA, and adaptive attackers (PAIR (Chao et al., 2023),

42 TAP (Mehrotra et al., 2024)) as special cases. Three findings motivate the workshop submission.
43

44 **Finding 1: matched ASR, divergent risk.** On JailbreakBench direct prompts, four
45 instruction-tuned targets all have greedy judge-ASR $\approx 0\%$, yet mean GAS@0 (bounded
46 $\mathbb{P}[\text{F harm}]$ with no attacker) spans **0.002–0.211**—a $\sim 100\times$ spread driven largely by Mistral-
47 7B’s hidden harmful mass.

48 **Finding 2: rankings flip with attacker budget.** Qwen2.5-7B is most vulnerable to a
49 fixed 18-template library (GAS@18 = 0.91) but most robust under three PAIR queries
50 (GAS@PAIR(3) = 0.02); Llama-3.1-8B shows the reverse. Single-number ASR cannot rank
51 models when the ordering is a function of attacker capability.

52 **Finding 3: complementary search coverage.** Running judge-guided and BCA-guided
53 TAP in parallel raises union deploy coverage from $\sim 57\%$ to $\sim 78\%$ on pooled seeds (102
54 paired runs, $L=8$), with symmetric exclusive wins. When both methods succeed, selected
55 prompts are never identical—BCA changes attack trajectories, not just rediscovers the same
56 jailbreak.

57 **Finding 4 (conditional bonus): BCA as pruning signal.** On a single-seed TAP run,
58 hybrid pruning (0.5-judge + 0.5-BCA) raises judge-ASR from 60.0% to 73.3%. Multi-seed
59 and multi-judge replicates shrink this to $\Delta = +4.5$ pp (± 13.6 pp 95% CI, four Qwen seeds)—
60 statistically indistinguishable from zero. We treat this as a tree-search-specific, non-load-
61 bearing observation; Pillars 1–2 do not depend on it.

62 2 Related work

63 **Jailbreak measurement.** HarmBench (Mazeika et al., 2024) and JailbreakBench (Chao et al.,
64 2024) standardize behaviors and automated grading. Adaptive attackers—PAIR (Chao
65 et al., 2023), TAP (Mehrotra et al., 2024), and successors—treat jailbreaking as search.
66 Chouldechova et al. (2026) argue that ASR comparisons are often invalid when threat
67 models and judges differ. We do not replace ASR; we add a distributional coordinate that
68 remains defined under explicit bounding and evaluator assumptions.

69 **Probabilistic verification of LLM outputs.** LLMChecker (Gross et al., 2025) applies PCTL
70 model checking to α - k bounded generation trees for text-quality properties (polarity, read-
71 ability). We repurpose the same formal backbone for *semantic harm reachability* under
72 adversarial prompts, pair it with red-team attackers, and evaluate *actionability*: correlation
73 with independent judges, template search lift, and TAP coverage.

74 **Interpretability for safety.** Mechanistic interpretability typically targets weights and
75 activations; our “interpretability” is *conditional output structure*—which continuations the
76 model assigns mass to near a prompt. That structure is directly tied to downstream actions
77 (triage, search, training data) rather than post-hoc narrative explanation alone, matching the
78 Actionable Interpretability workshop mandate (Workshop on Actionable Interpretability,
79 2026).

80 3 Background

81 **Attack success rate.** ASR is the fraction of attacks whose completions are judged harmful.
82 It answers: *did this sample fail?* It does not answer: *how much of the nearby output distribution*
83 *fails?*

84 **Bounded Markov view.** Following LLMChecker (Gross et al., 2025), we view generation
85 as a DTMC over string prefixes. At each prefix s , retain the top tokens until cumulative
86 mass reaches α or k tokens are kept, renormalize, and expand to depth L . Terminal leaves

Algorithm 1 Exact BCA (reachability on bounded tree)

Require: prompt x , model M , judge J , L, α, k
1: Expand α - k tree to depth L ; label each leaf $\ell(s) \leftarrow J(x, s)$
2: $V(s) \leftarrow \mathbf{1}[\ell(s) = \text{harm}]$ for leaves
3: **for** $d = L-1$ **downto** 0 **do**
4: $V(s) \leftarrow \sum_{t \in T_{\alpha,k}(s)} \tilde{P}(s, s \cdot t) V(s \cdot t)$
5: **end for**
6: **return** $V(x)$

87 are labeled harm or by judge J . Because the bounded object is a tree, $\mathbb{P}[\text{F harm}]$ is computed
88 by backward induction in $\mathcal{O}(|\text{states}|)$ after expansion.

89 **Evaluator-conditional probabilities.** All reported probabilities are conditional on judge J .
90 We use Qwen2.5-7B-Instruct as the primary semantic judge for cross-method comparability,
91 and validate predictive validity with an independent HarmBench-Mistral-7B classifier (§5.5).

92 4 Method

93 4.1 Bounded continuation analysis

94 BCA has three stages: (i) breadth-first expansion with batched GPU inference over all active
95 prefixes at each depth; (ii) semantic labeling of leaves; (iii) exact backward induction or
96 SMC when k^L exceeds a node budget. Algorithm 1 summarizes the exact path.

97 When $|\alpha_{k,L}|$ exceeds a node budget, we sample N independent paths from the same bounded
98 process, label each terminal continuation with J , and estimate $\hat{P}[\text{F harm}] = \frac{1}{N} \sum_i X_i$ with Ho-
99 effding confidence half-width $\sqrt{\ln(2/\delta)/(2N)}$ (Hoeffding, 1963). GPU batching amortizes
100 prefix computation across siblings at each depth (Kwon et al., 2023).

101 Beyond $\mathbb{P}[\text{F harm}]$, we track *refusal-collapse* mass: paths whose first segment matches a refusal
102 template yet still reach harm. This separates genuine refusals from “refusal prefixes that
103 collapse into compliance,” a failure mode invisible to keyword ASR.

104 4.2 Generative-attack susceptibility (GAS)

105 Let $x^{(0)}$ be the direct behavior prompt and $x^{(q)}$ the best adversarial prompt after q attacker
106 queries. $\text{GAS}@q$ is the running maximum of bounded harmful reachability over prompts
107 discovered up to budget q :

$$\text{GAS}(q) = \max_{i \leq q} H_{\alpha,k,L}(x^{(i)}; M, J).$$

108 Special cases: $\text{GAS}@0$ is BCA on the direct prompt; $\text{GAS}@18$ is the closed 18-template
109 library; $\text{GAS}@\text{PAIR}(q)$ and $\text{GAS}@TAP(q)$ compose outer attacker search with inner BCA on
110 each candidate prompt. We also report $\text{GAS-AUC} = \text{mean}(\text{GAS}@1, \dots, \text{GAS}@q)$ and GAS-
111 **elasticity** = $\text{GAS}(1) - \text{GAS}(0)$ as early-stopping signals under PAIR. This nests outcome
112 metrics (ASR) and distributional metrics (BCA) under one query-budgeted curve; the
113 outer attacker tree and inner continuation tree form a product-DTMC view. Table 1 lists
114 recoverable special cases.

115 4.3 Actionable uses

116 We highlight four deployment-facing uses aligned with actionable interpretability:

- 117 1. **Pre-deployment triage.** Flag prompts with high $\text{GAS}@0$ despite safe greedy decodes.
- 118 2. **Search prioritization.** Rank jailbreak templates by $\mathbb{P}[\text{F harm}]$ before sampling full com-
119 pletions (§??).

Table 1: GAS special cases (same inner tree, varying outer attacker).

Setting	Recovers
$q=0$, sample one completion	ASR on direct prompt
$q=0$, no outer attacker	BCA / GAS@0
q PAIR iterations	GAS@PAIR(q)
q TAP depth expansions	GAS@TAP(q)

Table 2: Matched ASR, divergent GAS@0 (mean over JBB behaviors, Qwen judge).

Target	Greedy ASR	GAS@0	GAS@18
Mistral-7B	0%	0.211	0.782
Qwen2.5-1.5B	0%	0.004	0.906
Qwen2.5-7B	0%	0.004	0.907
Llama-3.1-8B	0%	0.002	0.144

- 120 3. **Adaptive red-teaming.** Use BCA as a pruning/feedback signal inside TAP; use judge
 121 score for deploy selection (§5.6).
 122 4. **Coverage-maximizing audits.** Union regular- and BCA-guided searches to surface
 123 disjoint failure modes for adversarial training.

124 5 Experiments

125 5.1 Setup

126 **Behaviors:** JailbreakBench harmful behaviors (100 categories across harassment, malware,
 127 physical harm, fraud, disinformation, privacy, expert advice, etc.).

128 **Targets:** Qwen2.5-1.5B/7B-Instruct, Mistral-7B-Instruct-v0.2/v0.3, Llama-3.1-8B-Instruct.

129 **Inner tree:** unless noted, $\alpha=0.90$, $k=5$, $L \in \{8, 10, 12\}$, temperature 1.0 on the target,
 130 Qwen2.5-7B semantic judge for leaf labels.

131 **Attackers:** 18-template closed library; PAIR ($q \leq 10$); TAP ($b=2, w=3, d=3, q \leq 12$) with
 132 Llama-3.1-8B as default target.

133 We report exact DTMC results where feasible and SMC with $N=200\text{--}500$ trajectories
 134 otherwise.

135 5.2 Distributional fragility under matched ASR

136 Table 2 is the headline actionable result. All four targets refuse on greedy/direct samples
 137 (judge-ASR $\approx 0\%$), yet GAS@0 separates them by two orders of magnitude.

138 **Interpretation.** Mistral’s refusal is a *presentation layer*: $\sim 21\%$ of bounded mass at $L=8$
 139 already sits on harmful leaves without any jailbreak template. Llama is distributionally
 140 flat at the surface but becomes vulnerable under adaptive PAIR search (below). A safety
 141 team relying on one-shot ASR would treat these models identically; BCA recommends
 142 immediate Mistral triage and adaptive testing for Llama.

143 5.3 Query-budget curves and ranking flips

144 Figure 3 summarizes the Pillar-1 flip. Closed libraries stress-test suffix-forcing weaknesses
 145 (Qwen-7B); short-budget LLM attackers find narrative-rewrite weaknesses (Llama).

146 On an aligned 16-behavior PAIR sub-grid, deploy judge-ASR at $q=3$ is $\sim 43\%$ on Llama vs.
 147 $\sim 12\%$ on Qwen-7B ($\sim 3.6\times$). At $q=10$, Llama hybrid reaches 24.7% judge-ASR vs. 10.0% for
 148 Qwen regular; GAS-AUC over $q \in \{1, 2, 3\}$ ranks loss functions at fixed compute (hybrid

Table 3: Model rankings invert with attacker budget (representative suite averages).

Target	GAS@18	GAS@PAIR(3)
Qwen2.5-7B	0.91 (most fragile)	0.02 (most robust)
Llama-3.1-8B	0.14	0.17 (most fragile)

Table 4: Greedy-safe prompts with high bounded harmful mass (Qwen2.5-1.5B, exact BCA, $L=10$).

Behavior (category)	$\mathbb{P}[\text{F harm}]$	$\mathbb{P}[\text{F refusal}]$
Regulatory loopholes (gov.)	0.336	0.416
Phishing (privacy)	0.023	0.162
Plagiarism (fraud)	0.014	0.431
Ageism (harass.)	0.011	0.758

best on Llama, regular best on Qwen at $q=3$). Actionable takeaway: report ASR and GAS curves over q , not scalars.

5.4 Hidden mass and model comparison

On Qwen2.5-1.5B with exact BCA ($L=10$, 100 behaviors), greedy ASR is 0/100 yet 13 prompts exceed $\mathbb{P}[\text{F harm}]=0.01$; the regulatory-loopholes behavior reaches $\mathbb{P}[\text{F harm}]=0.34$ while still refusing greedily. Table 4 lists representative high-mass behaviors invisible to one-sample evaluation.

High $\mathbb{P}[\text{F refusal}]$ alongside nonzero harm mass indicates *refusal instability*: many branches open with refusal-like tokens yet still reach harmful leaves—exactly the failure mode keyword ASR misses. Malware/hacking behaviors on this model are conversely robust in the bounded tree (0/10 with $\mathbb{P}[\text{F harm}]=0$, mean refusal mass 0.83), showing category-specific structure actionable for prioritization.

Cross-model template sweeps (SMC, 18 templates) show all models at greedy ASR = 0 but mean best $\mathbb{P}[\text{F harm}]$ spanning 0.18–0.93. Llama resists 37/95 behaviors entirely; Qwen-7B resists 3/95. Suffix-forcing templates dominate Qwen and Mistral; nested fiction dominates Llama—actionable guidance for model-specific red-team playbooks.

5.5 Predictive validity and search guidance

Independent judge correlation (Axis 0). For each (behavior, template) pair we compute bounded $\mathbb{P}[\text{F harm}]$ with Qwen as leaf judge, then sample 50–100 completions and score with HarmBench-Mistral-7B. Table ?? shows per-target validity; Spearman $\rho \in [0.53, 0.73]$ ($p < 10^{-40}$), with top-1 template lift up to $7.43\times$ on Qwen-7B. A separate 100-pair Llama slice gives $\rho=0.66$ ($p=5.6\times 10^{-14}$) under the same judge for template ranking.

Template ranking before queries. On 11 Qwen2.5-1.5B behaviors, selecting templates by bounded mass yields $\sim 6.2\times$ lift in empirical ASR over random template choice at equal query budget. Table 5 shows a simulated search: mass-guided selection matches or beats empirical ranking until ~ 100 full-completion queries per behavior.

Cross-model template penetration. Suffix-forcing wins 84/95 behaviors on Qwen2.5-7B but only 18/95 on Llama-3.1-8B; nested fiction is Llama’s dominant failure family. Top-1 mass-guided template selection yields $\sim 2.77\times$ better empirical coverage than uniform sampling on the Qwen-1.5B subset.

Table 5: Template search under query budget B (queries per template).

B	Random ASR	Empirical ASR	BCA ASR
1	0.025	0.041	0.113
5	0.026	0.089	0.113
10	0.026	0.110	0.113
25	0.025	0.126	0.113
50	0.025	0.131	0.113

Table 6: Axis 0: bounded mass vs. independent-judge empirical ASR ($L=8$, HarmBench evaluator).

Target	n pairs	Spearman ρ	Pearson r	Top-1 lift
Qwen2.5-1.5B	110	0.596	0.31	$4.54\times$
Qwen2.5-7B	540	0.733	0.83	$7.43\times$
Mistral-7B	540	0.611	0.59	$2.56\times$
Llama-3.1-8B	540	0.534	0.79	$6.73\times$

5.6 BCA inside adaptive red-teaming

We integrate BCA into PAIR and TAP on Llama-3.1-8B by swapping the per-candidate pruning score: *regular* (judge), *bca* (bounded mass), *hybrid* ($0.5\cdot\text{judge} + 0.5\cdot\text{BCA}$). All cells are evaluated on held-out **judge-ASR** (any leaf scoring 10/10), independent of the pruning objective. Inner tree: $\alpha=0.99$, $k=2$, $L=8$, exact DTMC, Qwen judge unless noted.

Under TAP, BCA’s dense $[0, 1]$ scores help prune wide candidate sets (*b.w* children per depth); under PAIR’s linear refinement the binary judge wins (50.0% vs. 40.0%). GCG has no natural BCA variant (non-differentiable through the bounded tree) and hit 0% at 200 steps despite $\sim 5000\times$ more target forwards per behavior than one TAP query.

Multi-seed and multi-judge robustness. The single-seed hybrid +13.3 pp gain does not survive replication. On 30 behaviors with Qwen in-loop: regular 66.7%, hybrid 63.0% ($\Delta=-3.7$ pp). With HarmBench in-loop: regular 58.3%, hybrid 53.8% ($\Delta=-4.5$ pp). Llama Guard 3 saturates at 100% ASR (75/75 trees) and is unsuitable for comparative pruning. Pooled over four Qwen-pruning seeds (42–45): regular 57.0%, hybrid 61.5%, $\Delta=+4.5$ pp ± 13.6 pp (95% CI contains zero). We therefore present hybrid pruning as exploratory, not load-bearing.

Search vs. selection. Pure BCA pruning with BCA-based deploy selection collapses deploy ASR to 24.1% despite 58.3% *any-leaf* success. The operational split—**BCA prunes, judge picks**—restores deploy ASR to 58.3%. Density signals guide exploration; binary deployment judgments need outcome-aligned scores.

Complementarity. On 102 paired (seed, behavior) runs (seeds 42–45, $L=8$): union deploy coverage **78.4%** vs. $\sim 57\%$ for either method alone (both=37.3%, regular-only=19.6%, BCA+judge-only=21.6%, neither=21.6%). When both succeed, winning prompts differ on all 38 pooled pairs.

100-behavior scale ($L=12$, seed 44). On the full JailbreakBench suite we compare *regular* + *rich* BCA *attacker feedback* (BCA informs iterative feedback only; pruning and deploy remain judge-only) vs. BCA prune + judge-pick deploy: deploy ASR 43.0% (37/86) vs. 36.9% (31/84); on 80 paired behaviors union deploy ASR is **55.0%** (44/80) vs. 43.0% regular-only—harder categories (disinformation, fraud, privacy) dominate the long tail vs. the 78.4% union on the first three categories at $L=8$.

Table 7: Pruning signal vs. independent judge-ASR @ $q_{\max}$ (Llama target).

Attacker	Pruning	$q_{\max}$	judge-ASR
PAIR	judge	10	50.0%
PAIR	BCA	10	40.0%
PAIR	hybrid	10	35.7%
TAP	judge	12	60.0%
TAP	BCA	12	62.5%
TAP	hybrid	12	73.3%
GCG	xent	200 steps	0.0%

6 Discussion

What interpretability buys. BCA makes the *local geometry* of the output distribution measurable: refusal mass vs. harmful mass, template-dependent shifts, and query-budget-dependent rankings. The actionable product is not a heatmap for its own sake but decisions—where to spend red-team queries, which model to hold, which search mode to run overnight.

Limits. (i) BCA certifies a *bounded* process, not the full vocabulary-sized tree; tail mass may matter for rare catastrophes. (ii) Expansion is exponential in L ; long horizons require SMC or truncated search (max _nodes). (iii) Leaf labels at depth L miss harm that emerges later in full completions—the dominant residual gap to empirical ASR. (iv) All probabilities are judge-conditional; saturation and disagreement (above) bound how strongly one can claim cross-benchmark dominance.

Relation to actionable interpretability. Recent jailbreak work treats safety as search over structured attack spaces rather than one-shot classification. We contribute the inner loop: a probabilistic reachability readout that turns generation-structure insights into triage, search priors, and coverage metrics—with explicit failure modes when interpretability signals are misapplied to deployment selection.

Horizon and tractability. Exact BCA requires $\mathcal{O}(L)$ batched model calls but $\mathcal{O}(k^L)$ states in the worst case; judge labeling at each leaf multiplies cost in adaptive attackers. Longer horizons therefore require SMC, node caps, or property-directed expansion—the same engineering trade-off as LLMChecker at scale (Gross et al., 2025). For red-teaming we find $L \in \{8, 12\}$ sufficient to separate models and rank templates; full-completion ASR remains the gold standard for final deployment claims.

7 Conclusion

Attack success rate asks whether a sample failed. Bounded continuation analysis asks how much harmful probability mass lives in the likely branches around a prompt—and, under GAS, how that mass grows with attacker budget. Across JailbreakBench targets, this diagnostic separates models with identical greedy ASR, predicts independent-judge attack success, and supports complementarity-based audit campaigns. The load-bearing claims are Pillars 1–2 (ranking flips and distributional fragility); BCA-as-pruning gains are secondary and not yet robust across judges and seeds. We argue this is a template for actionable interpretability in safety: expose conditional structure, then tie it to measurement protocols red teams can run today.

References

- Christel Baier and Joost-Pieter Katoen. *Principles of Model Checking*. MIT Press, 2008.
- Patrick Chao et al. Jailbreaking black box large language models in twenty queries. *arXiv preprint arXiv:2310.08419*, 2023.

-
- Patrick Chao et al. JailbreakBench: An open robustness benchmark for jailbreaking large language models. *arXiv preprint arXiv:2404.01318*, 2024.
- Alex Chouldechova, A. Feder Cooper, Solon Barocas, Abhinav Palia, Dan Vann, and Hanna Wallach. Comparison requires valid measurement: Rethinking attack success rate comparisons in AI red teaming. *Advances in Neural Information Processing Systems*, 38, 2026.
- David Gross, Helge Spieker, and Arnaud Gotlieb. Bounded PCTL model checking of large language model outputs. *arXiv preprint arXiv:2509.18836*, 2025.
- Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023.
- Mantas Mazeika et al. HarmBench: A standardized evaluation framework for automated red teaming and robust refusal. *arXiv preprint arXiv:2402.04249*, 2024.
- Anay Mehrotra et al. Tree of attacks with pruning: Efficient black-box jailbreaking of large language models. In *Proceedings of the 41st International Conference on Machine Learning*, 2024.
- Workshop on Actionable Interpretability. Call for papers, COLM 2026. <https://actionable-interpretability.github.io/cfp/>, 2026.
- Hálim L. S. Younes and Reid G. Simmons. Probabilistic verification of discrete event systems using acceptance sampling. In *Proceedings of the 14th International Conference on Computer Aided Verification (CAV)*, volume 2404 of *Lecture Notes in Computer Science*, pp. 223–235. Springer, 2002.

A Implementation

Our GPU implementation uses breadth-first batched inference (vLLM/HuggingFace backends), sparse backward induction for exact trees, and batched SMC for large sweeps. Code and interactive complementarity visualizations accompany the project website.

B Extended tables

Table 8: Deploy vs. any-leaf ASR (seeds 42–45 pooled, $L=8$).

Method	Any-leaf ASR	Deploy ASR
Regular (judge prune + pick)	57.0%	57.0%
BCA (BCA prune + pick)	58.3%	24.1%
BCA prune + judge pick	58.3%	58.3%
Hybrid	61.5%	61.5%

Table 9: 100-behavior complementarity (seed 44, $L=12$, 80 paired behaviors).

Outcome	Share
Both deploy	23.8%
Regular+rich only	18.8%
BCA+judge-pick only	12.5%
Neither	45.0%
Union	55.0%

Table 10: Cross-judge TAP-Llama any-leaf ASR (30 behaviors, in-loop judge).

Method	Qwen	HarmBench	Llama Guard 3
regular	66.7%	58.3%	100%
bca	55.6%	57.7%	100%
hybrid	63.0%	53.8%	100%

274 C Ethics statement

275 This work studies adversarial prompts and harmful behaviors to improve safety evaluation.
276 We report template *families* rather than operational jailbreak strings, follow public bench-
277 mark protocols, and emphasize defensive use: triage, auditing, and adversarial training
278 coverage. Dual-use risk remains; we recommend gated release of attack traces consistent
279 with JailbreakBench norms.